Introduction & Review

→ Operator($\mathcal{L}$): a mapping b/w two fn. spaces

- $C^1(\mathbb{R})$ - space of cont. diff. fns on $\mathbb{R}$

eg: of operators: - $\mathcal{L} = \frac{d}{dx}$: $C^1(\mathbb{R}) \longrightarrow C(\mathbb{R})$
$u(x) \longrightarrow u'(x)$
[differentiation].

- $\mathcal{L} = \int_0^t$: $C(\mathbb{R}) \longrightarrow C^1(\mathbb{R})$
$u(x) \longrightarrow U(t) = \int_0^t u(x)\,dx$ [integ.]

→ Linear operator: satisfies $\mathcal{L}(\alpha u_1 + \beta u_2) = \alpha \mathcal{L}(u_1) + \beta \mathcal{L}(u_2)$,

α, β scalars & u_i any fns.

eg: $\mathcal{L} = \frac{d}{dx}$, $\mathcal{L} = \int_0^t$

What about - $\mathcal{L} = x^2 \frac{d}{dx}$?

$\mathcal{L}[\alpha u_1 + \beta u_2] = \alpha x^2 \frac{du_1}{dx} + \beta x^2 \frac{du_2}{dx} = \alpha \mathcal{L}[u_1] + \beta \mathcal{L}[u_2]$

⇒ linear

- $\mathcal{L} = \left(\frac{d}{dx}\right)^2$?

$\mathcal{L}[\alpha u_1 + \beta u_2] = \left[\alpha \frac{du_1}{dx} + \beta \frac{du_2}{dx}\right]^2 \neq \cdots$

→ Linear diff. eqn: If $\mathcal{L}$ is a diff. operator, then $\mathcal{L}u = f$ is a linear D.E where f is a known fn.

→ order of a D.E - highest value of its deriv.

→ Homogeneity - a linear eqn of the form $\mathcal{L}u = 0$

i.e, $u = 0$ is a sol. [else non homog.]

→ Principle of superpos - If u_1, u_2 are sol. of a linear homog eqn, $\alpha u_1 + \beta u_2$ is also a sol.

Review of ODEs.

$$\frac{du}{dt} = f(t,u) \quad ; \quad \underbrace{u(0) = u_0}_{\text{initial cond.}}$$

eg: $u' + \lambda u = 0$, $\lambda \in \mathbb{R}$

$u(0) = u_0$

→ first order, linear, homog. ODE

sol: $u(t) = u_0 e^{-\lambda t}$.

eg: $u'' + \lambda u = 0$, $u(0) = u_0$, $u'(0) = V_0$, $\lambda \in \mathbb{R}$.

Case 1 - $\lambda = 0$: $u'' = 0$, $u(0) = u_0$, $u'(0) = V_0$

$\rightarrow u(x) = u_0 + V_0 x$

Case 2 - $\lambda \neq 0$: try $u(x) = e^{\gamma x}$

$\Rightarrow$ Need $\gamma^2 + \lambda = 0 \Rightarrow \gamma = \pm\sqrt{-\lambda} = \gamma_1, \gamma_2$

$\Rightarrow u(x) = C_1 e^{\gamma_1 x} + C_2 e^{\gamma_2 x}$, C_j det. based on I.C.

Case 2a). $\lambda > 0$: $\gamma^2 = -\lambda$ (<0) $\Rightarrow$ roots are $\pm i\sqrt{\lambda}$.

$\Rightarrow u(x) = C_1 e^{i\sqrt{\lambda}x} + C_2 e^{-i\sqrt{\lambda}x}$

$= \underbrace{(C_1 + C_2)}_{A} \cos\sqrt{\lambda}x + \underbrace{i(C_1 - C_2)}_{B} \sin\sqrt{\lambda}x$

$A = u_0$, $B = \frac{V_0}{\sqrt{\lambda}}$.

Case 2b). $\lambda < 0$: $\gamma^2 = -\lambda$ $(>0) \Rightarrow \gamma = \pm\sqrt{-\lambda} = \pm\omega$.

& $u(x) = C_1 e^{\omega x} + C_2 e^{-\omega x}$.

Note: $\cosh(x) = \frac{e^x + e^{-x}}{2}$; $\sinh(x) = \frac{e^x - e^{-x}}{2}$.

[$\Rightarrow \cosh'(x) = \sinh(x)$; $\sinh'(x) = \cosh x$]

$\Rightarrow u(x) = A\cosh(\omega x) + B\sinh(\omega x)$,

$A = u_0$ & $B = \frac{V_0}{\omega}$.

Partial Differential Equations (PDEs).

→ PDE is an eqn for a fn of $n \geq 2$ variables, involving its partial deriv.

eg: $u : \mathbb{R} \times [0,\infty) \longrightarrow \mathbb{R}$

$u(x,t)$: $\partial_t u + \partial_x u = 0$. [first order, linear homog.]

sol: Try $u(x,t) = f(x-t)$.

eg: $\partial_x u = 2x \sin y + e^{xy}$. ["]

sol: integrate wrt x:

$\rightarrow u(x,y) = x^2 \sin y + \dfrac{e^{xy}}{y} + C(y)$

eg: $\partial^2_{yx} u = 0$. (2nd order, ")

sol: integrate first wrt x & then wrt y.

$\rightarrow \partial_y u = C(y)$, $\Rightarrow u(x,y) = G(y) + F(x)$

$G'(y) = C(y)$.

eg: $\partial_t u = \partial^2_x u$ (").

Sol: Try $u(x,t) = 2at + bx + ax^2$

→ This works, but not a gen. sol.

Four Fundamental PDEs.

1. Laplace eqn: $u = u(x,y)$: $\partial^2_x u + \partial^2_y u = 0$.

Notation: $\Delta u = 0$

[$\Delta u = f \neq 0$ is the Poisson eqn]

→ arises in electrostatics, steady state heat eqn, etc.

2. Heat eqn : $u = u(t, x)$

$$\partial_t u = \partial_x^2 u$$

→ heat conduction, diffusion process, etc

3. Wave eqn : $u = u(t, x)$

$$\partial_t^2 u = \partial_x^2 u$$

4. Schrödinger eqn : $u = u(t, x)$

$$\partial_t u = i \partial_x^2 u$$

[arises in quantum mech.]

eg. of non-linear PDEs

→ Eikonal eqn : $(\partial_x u)^2 + (\partial_y u)^2 = \frac{\omega^2}{c^2}$

(arises in optics)

→ Korteweg-De Vries eqn (KdV) : $\partial_t u - 6u\partial_x u + \partial_x^3 u = 0$

(water waves)

Some notation

→ grad. operator ∇ : $\nabla f(x_1, \ldots, x_n) = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \ldots, \frac{\partial f}{\partial x_n}\right)$

→ divergence operator : $\nabla \cdot \underset{(F_1, \ldots F_n)}{\underset{\sim}{\vec{F}}} = \sum_{j=1}^{n} \frac{\partial F_j}{\partial x_j}$

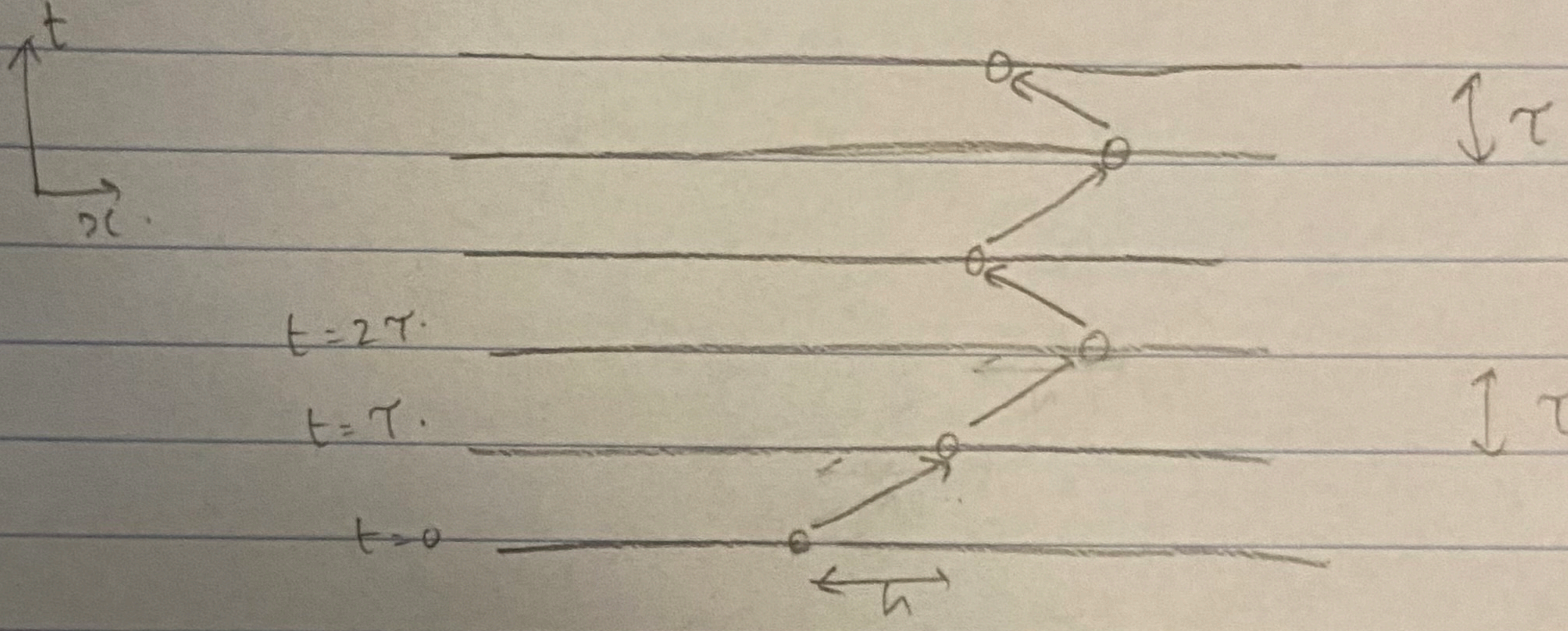
→ Laplace operator / Laplacian : $\nabla \cdot \nabla = \nabla^2 = \Delta$

$$\Delta f = \sum_{j=1}^{n} \partial_{x_j}^2 f$$

Heat equation

$$\underbrace{u(t,x)}_{\text{temperature}} : \partial_t u = K \partial_{xx} u + \underbrace{Q}_{\text{source}} \quad , \quad t > 0, \quad x \in (0, L)$$

motivation: Random walk, heat conduction



$$p(t+\tau, x) = \frac{1}{2} p(t, x-h) + \frac{1}{2} p(t, x+h)$$

$$p(t,x) + \tau \partial_t p\big|_{(t,x)} + O(\tau^2) = \frac{1}{2}\Big[p(t,x) - h\partial_x p\big|_{(t,x)} + \frac{h^2}{2}\partial_x^2 p\big|_{(t,x)}$$

$$+ p(t,x) + h\partial_x p + \frac{h^2}{2}\partial_x^2 p\Big] + O(h^3)$$

$$\Rightarrow \underset{+O(\tau^2)}{\tau \partial_t p} = h^2 \partial_x^2 p + O(h^3)$$

$$\Rightarrow \partial_t p = \underbrace{\left(\frac{h^2}{\tau}\right)}_{K = O(1)} \partial_x^2 p + O\left(\frac{h^3}{\tau}\right) + O(\tau)$$

$\lim h, \tau \to 0$

$$\Rightarrow \partial_t p = K \partial_x^2 p$$

$$\partial_t u = k\,\partial_x^2 u, \quad t>0, \quad x \in (0, L).$$
$$u(t=0, x) = f(x) \qquad [I \cdot C]$$

B.C: (i) prescribed temperatures (Dirichlet B.C.)

$$u(t, x=0) = T_1(t), \quad u(t, x=L) = T_2(t)$$

(ii) prescribed flux. (Neumann B.C.)

$$\partial_x u(t, x=0) = \phi_1(t), \quad \partial_x u(t, x=L) = ($$

→ When $\partial_x u(t, x=0) = \partial_x u(t, x=L) = 0$ → insulat bounda

(iii) Robin B.C

$$a_0\,\partial_x u(t, 0) + b_0\,u(t, 0) = \psi_0(t)$$
$$a_L\,\partial_x u(t, L) + b_L\,u(t, L) = \psi_L(t).$$

Higher dimensions ($x \in \mathbb{R}^d$)

$$\partial_t u = \nabla \cdot (K(x)\nabla u) + Q$$

For constant k, $\partial_t u = k\Delta u + Q$

Laplacian in various coordinates

→ cartesian ($\mathbb{R}^3$): $\Delta u = \partial_x^2 u + \partial_y^2 u + \underbrace{\partial_z^2 u}_{\text{absent in 2D}}$

→ polar coord: (r, θ), $x = r\cos\theta$, $y = r\sin\theta$

Ex. PT: $\Delta u(r,\theta) = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2}.$

→ cylindrical coord: (r, θ, z)

$$\Delta u = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u}{\partial\theta^2} + \frac{\partial^2 u}{\partial z^2}$$

→ spherical coord: (r, θ, ϕ), $x = r\sin\phi\cos\theta$

$y = r\sin\phi\sin\theta$, $z = r\cos\phi$

Ex: PT $\Delta u(r,\theta,\phi) = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial u}{\partial r}\right) + \frac{1}{r^2\sin\phi}\frac{\partial}{\partial\phi}\left(\sin\phi\frac{\partial u}{\partial\phi}\right) + \frac{1}{r^2\sin^2\phi}\frac{\partial^2 u}{\partial\theta^2}$